

European Bioinformatics Institute, Wellcome Genome Campus, Hinxton, United Kingdom

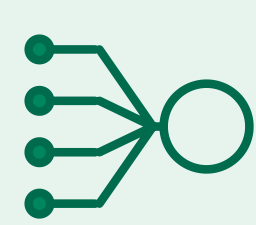


A single pooled genome-scale Perturb-Seq dataset can **simultaneously profile tens of millions of cells across hundreds of thousands of perturbations**. That scale could reveal robust biology across studies, but identifiers, metadata, assays and processing pipelines differ. The Perturbation Catalogue aims to unify datasets across modalities and provide a shared access path for this evidence.

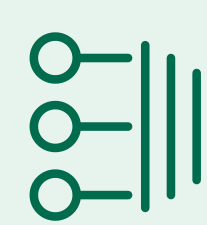
Year	Number of people in the labour force (millions)
2005	18
2025	25

Europe PMC title/abstract query; 24 Aug 2026.
Scale example: *FirstWord Pharma*, story 7139212.

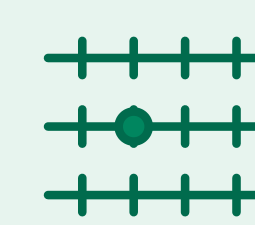
169 diseases



Pooled guides → gene-level phenotypes and fitness.



Guide identity + scRNA-seq → cell-state responses.



Variant libraries + assays → sequence-function maps.

Organism, tissue, cell line, disease

Assay, technology, kit, reference

Produce the final analysis-ready H5AD

Five Perturb-Seq datasets are live. Perturbation-control comparisons feed DEA and Hallmark GSEA. More datasets and modalities coming soon.

[ebi.ac.uk/perturbation-catalogue](#)

Perturbation Catalogue

Targets Datasets API Documentation About Request dataset

Perturbation Catalogue

① Targets ▾ Search by metadata fields 🔍 Search

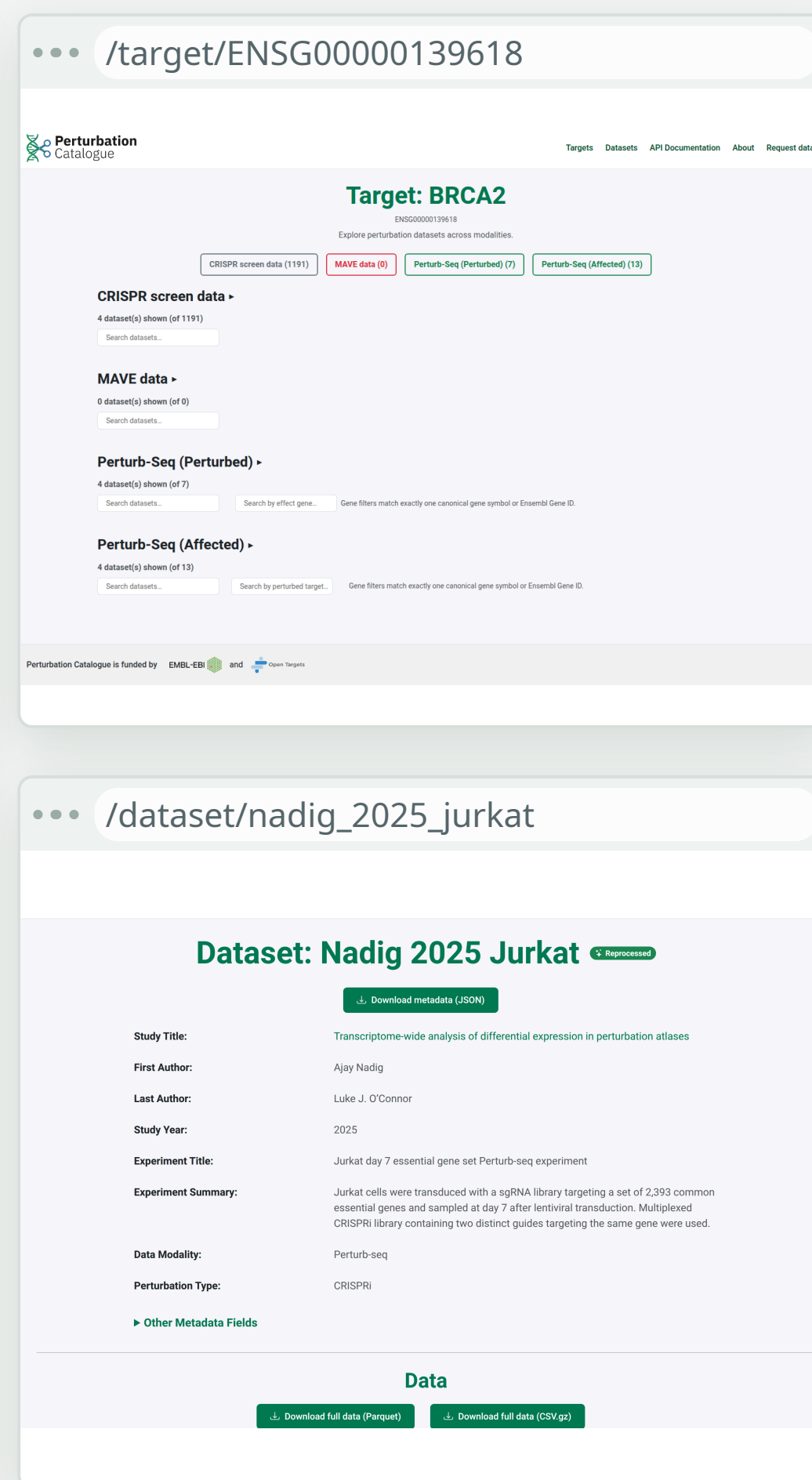
Try searching for:

Perturbation Catalogue is a curated database that brings together data from various genetic perturbation experiments, including PerturbSeq, CRISPR and MAVER screens, making it easier for researchers to study how modifying genes or proteins affects biological function across various biological and molecular contexts.

Perturbation Catalogue at a glance

Explore high-level statistics across the catalogue before diving into specific targets.

DATASETS 1237 (spanning 2014 - 2025)	TARGETS 19235	UNIQUE TISSUES 36	UNIQUE CELL TYPES 34
UNIQUE CELL LINES 1200	UNIQUE DISEASES 169		



```
{
  "perturbation": {"gene_symbol":
"TFAM"},
  "effect": {"gene_symbol": "MT-ND5",
"log2fc": -1.926, "padj": 0.0}
}
```

Data is available in **Parquet** and **CSV.gz**, and metadata as **JSON**.

AI-assisted drafting with curator review and validated ontology mapping.



More kits, experimental structures and independent quality metrics.



Explicit provenance and stable snapshots for reproducible downstream use.



Enrich variant-to-function coverage with harmonised scores.

Visit: ebi.ac.uk/perturbation-catalogue Contact: perturbation-catalogue-help@ebi.ac.uk

